Skills Development for Adolescent Girls

Investment case · East Java, Indonesia

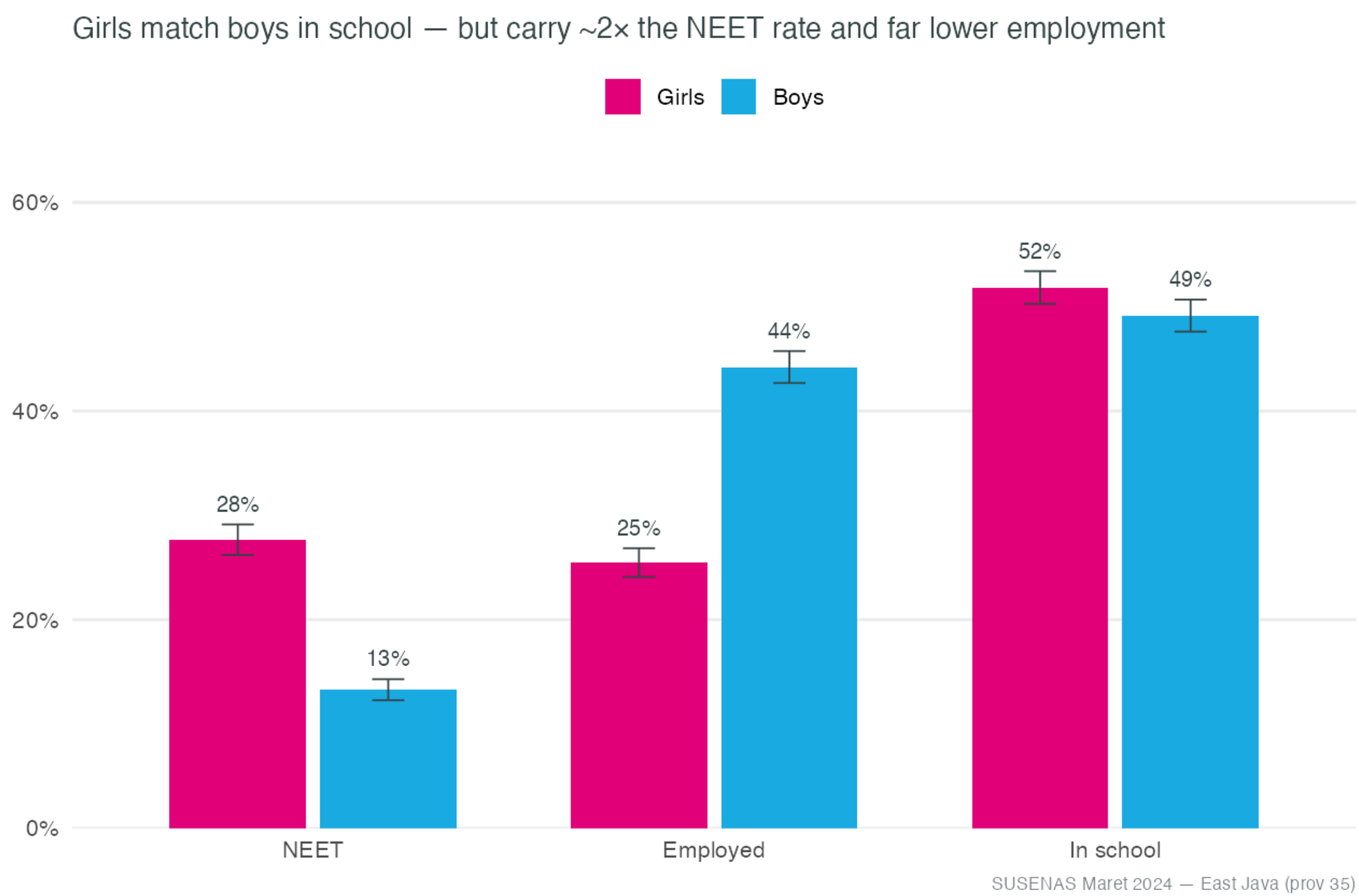
Hernando Grueso, PhD · UNICEF

2026-07-01

Contents

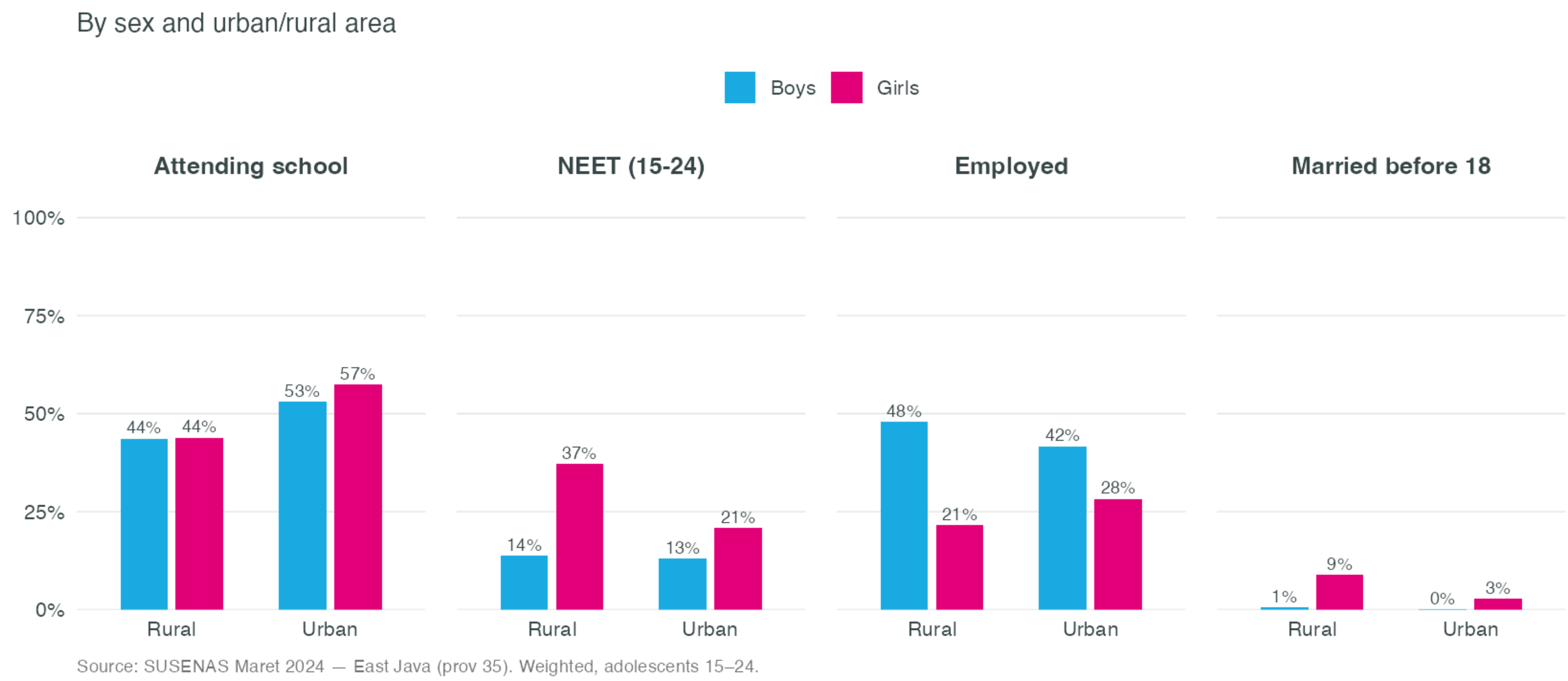
1 · Diagnostic — scale & profile of the gap

The gap is in the transition, not the classroom

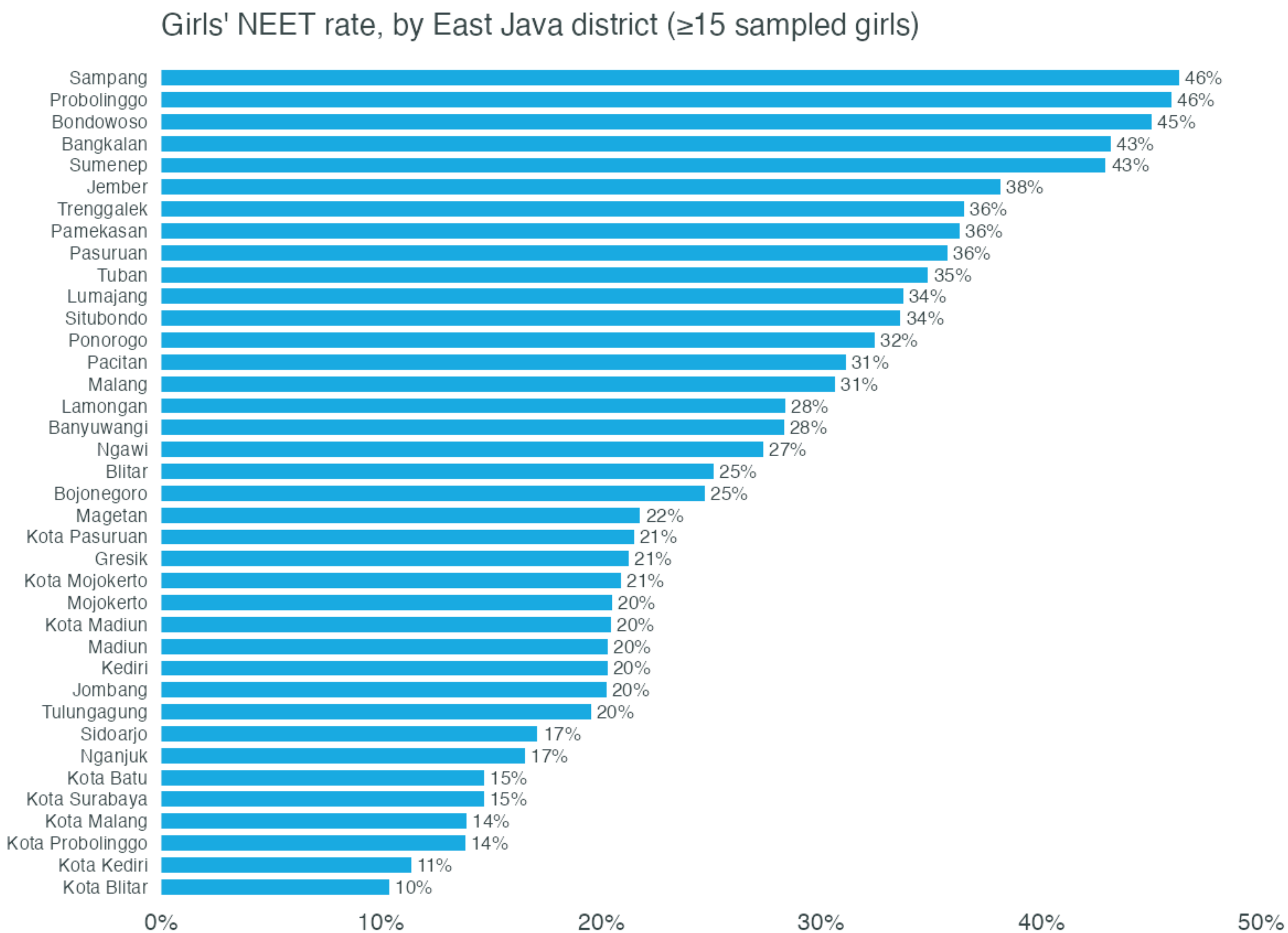


Girls 15–24 are at parity (slightly ahead) in school, yet their NEET rate is **~2× boys’** (28% vs 13%) and employment far lower (26% vs 44%) — the disadvantage is the school-to-work transition (marriage, caregiving, gender norms), not schooling. SUSENAS Maret 2024, survey-weighted; 95% CIs.

Outcomes by area of residence

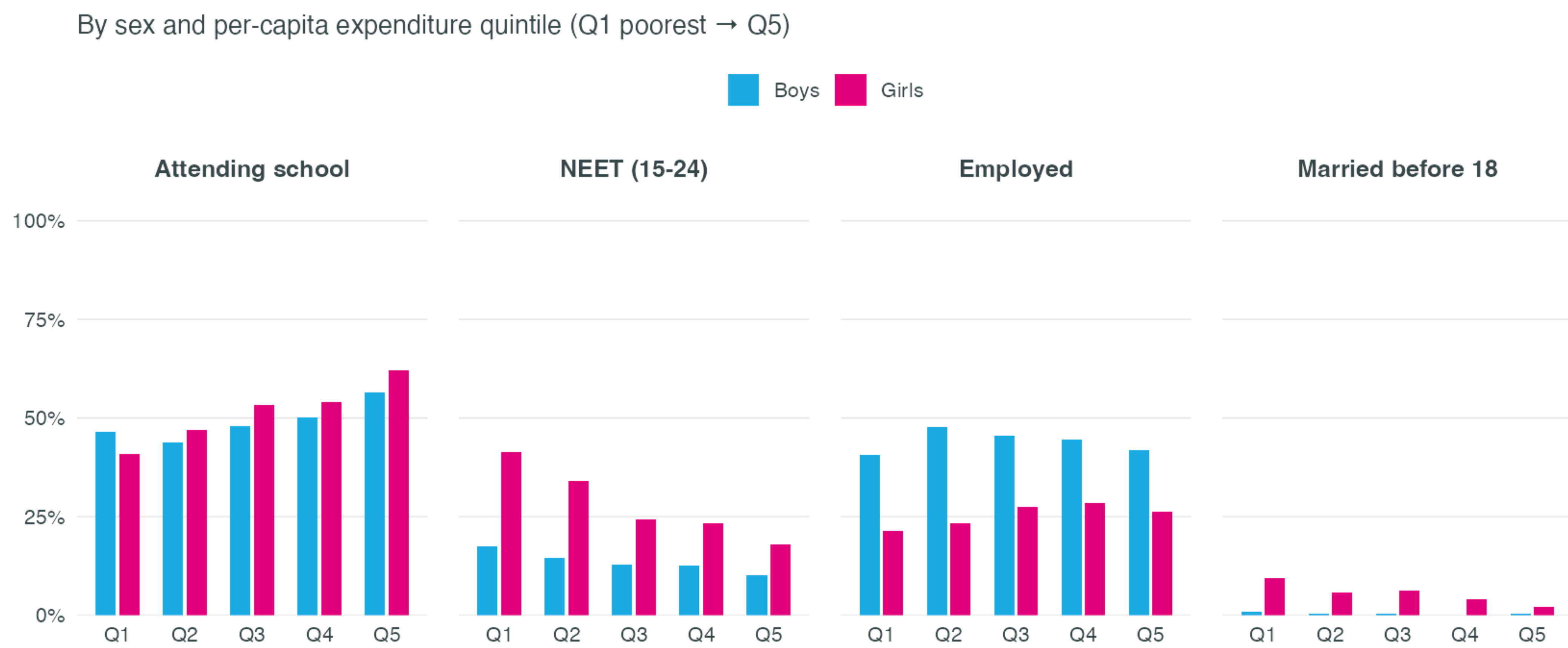


Where NEET concentrates — by district



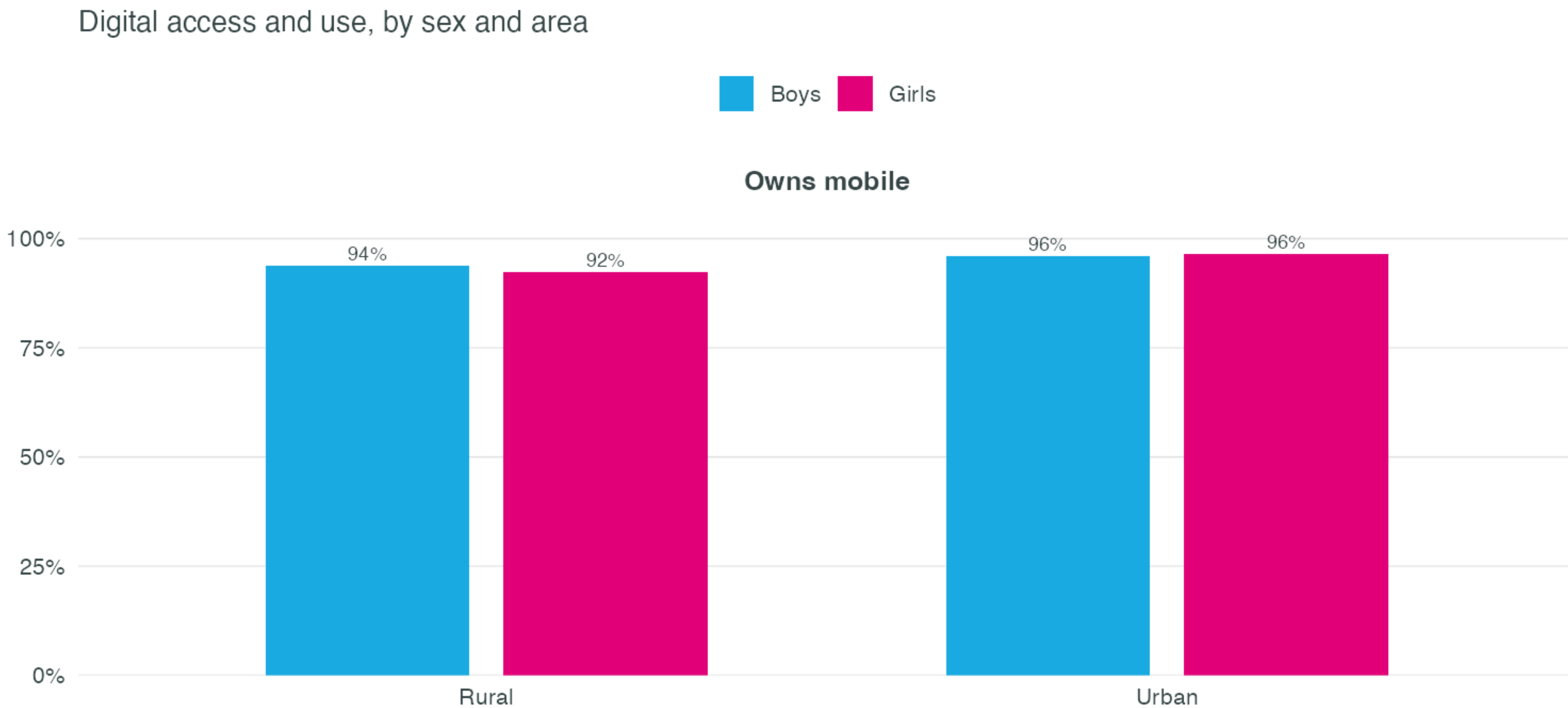
Source: SUSENAS Maret 2024 — East Java (prov 35). Weighted, adolescents 15–24.

Outcomes by per-capita expenditure quintile



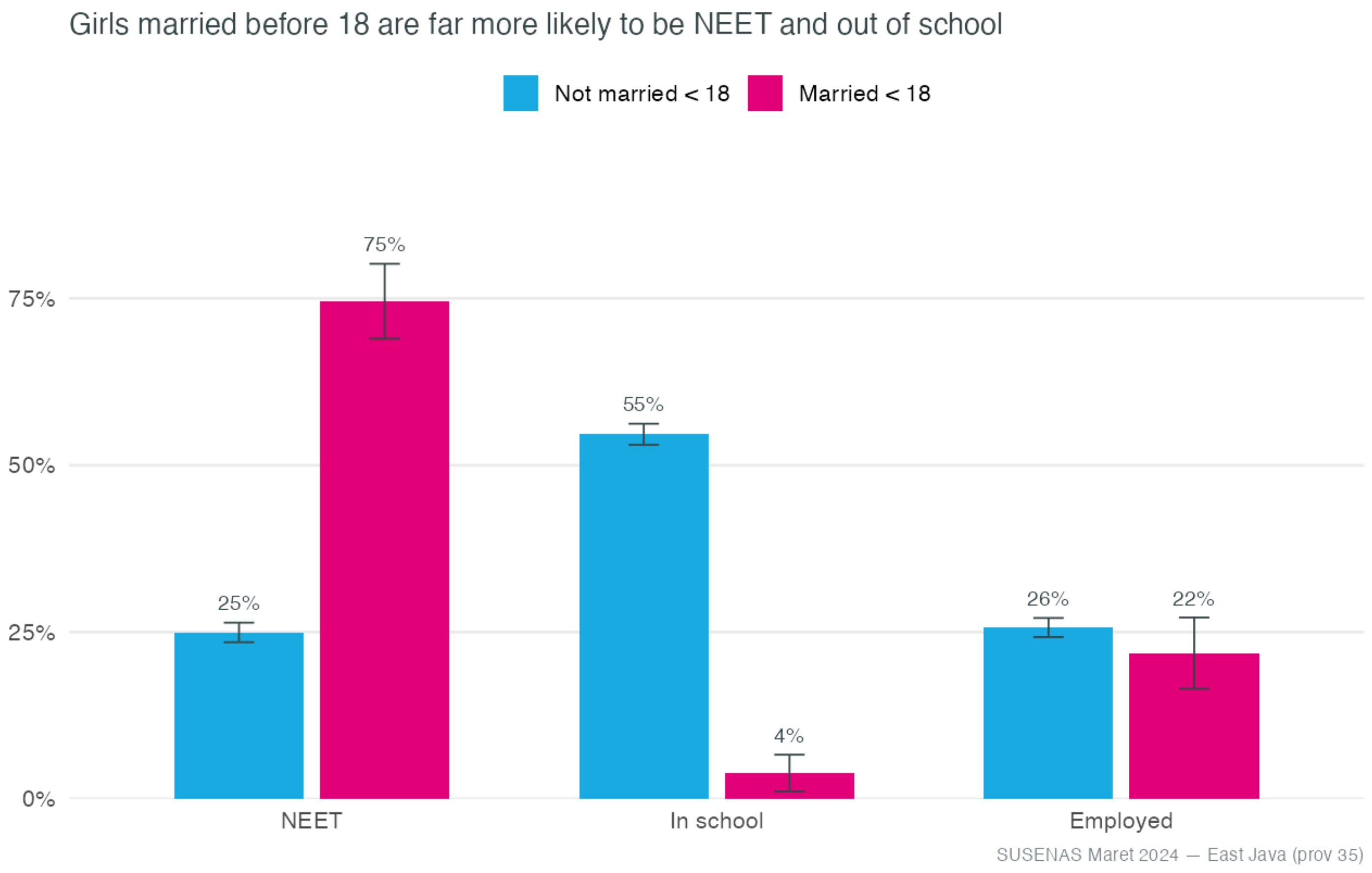
Source: SUSENAS Maret 2024 — East Java (prov 35). Weighted, adolescents 15–24.

The digital divide — access vs. use



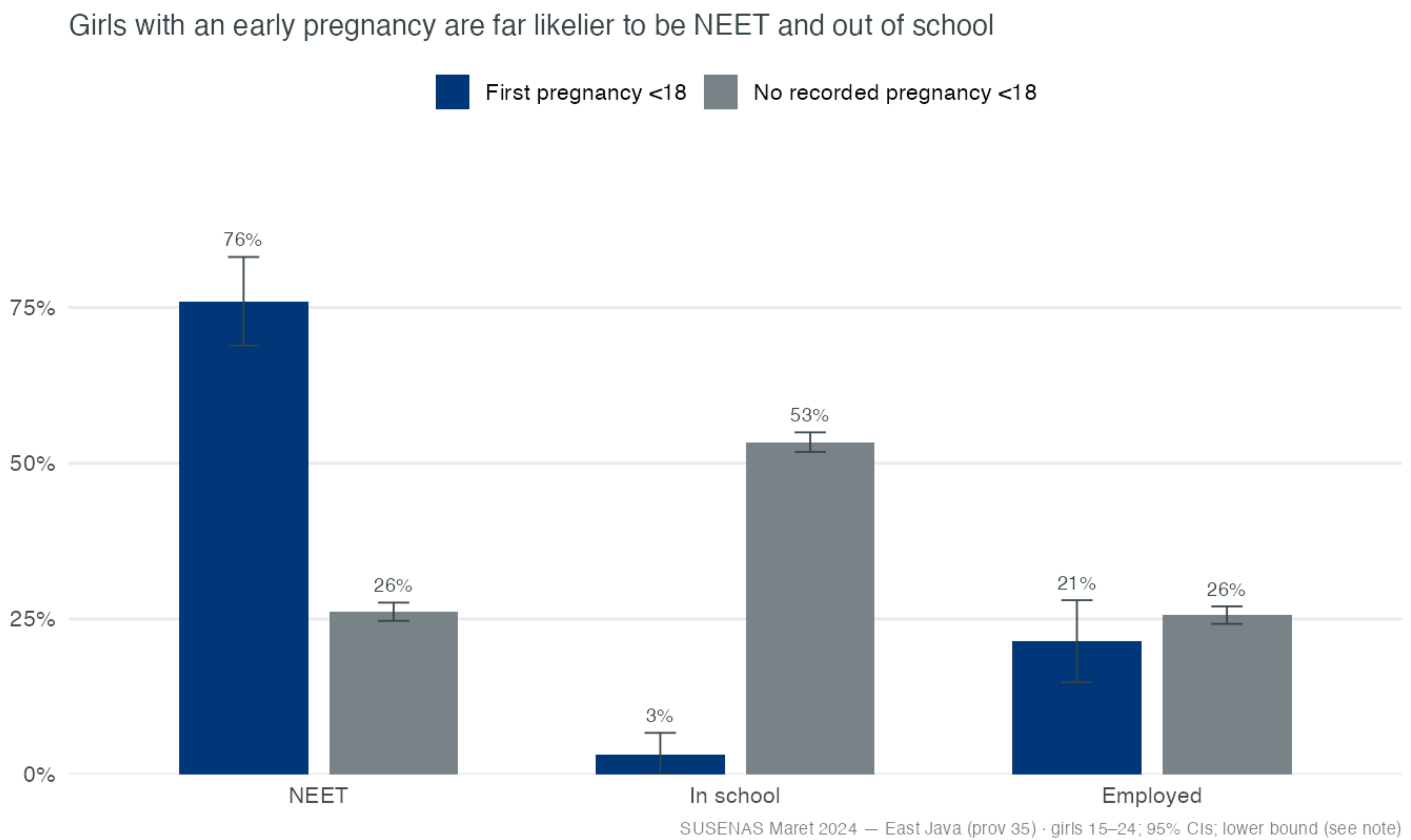
Source: SUSENAS Maret 2024 — East Java (prov 35). Weighted, adolescents 15–24.

Child marriage — a pathway into exclusion



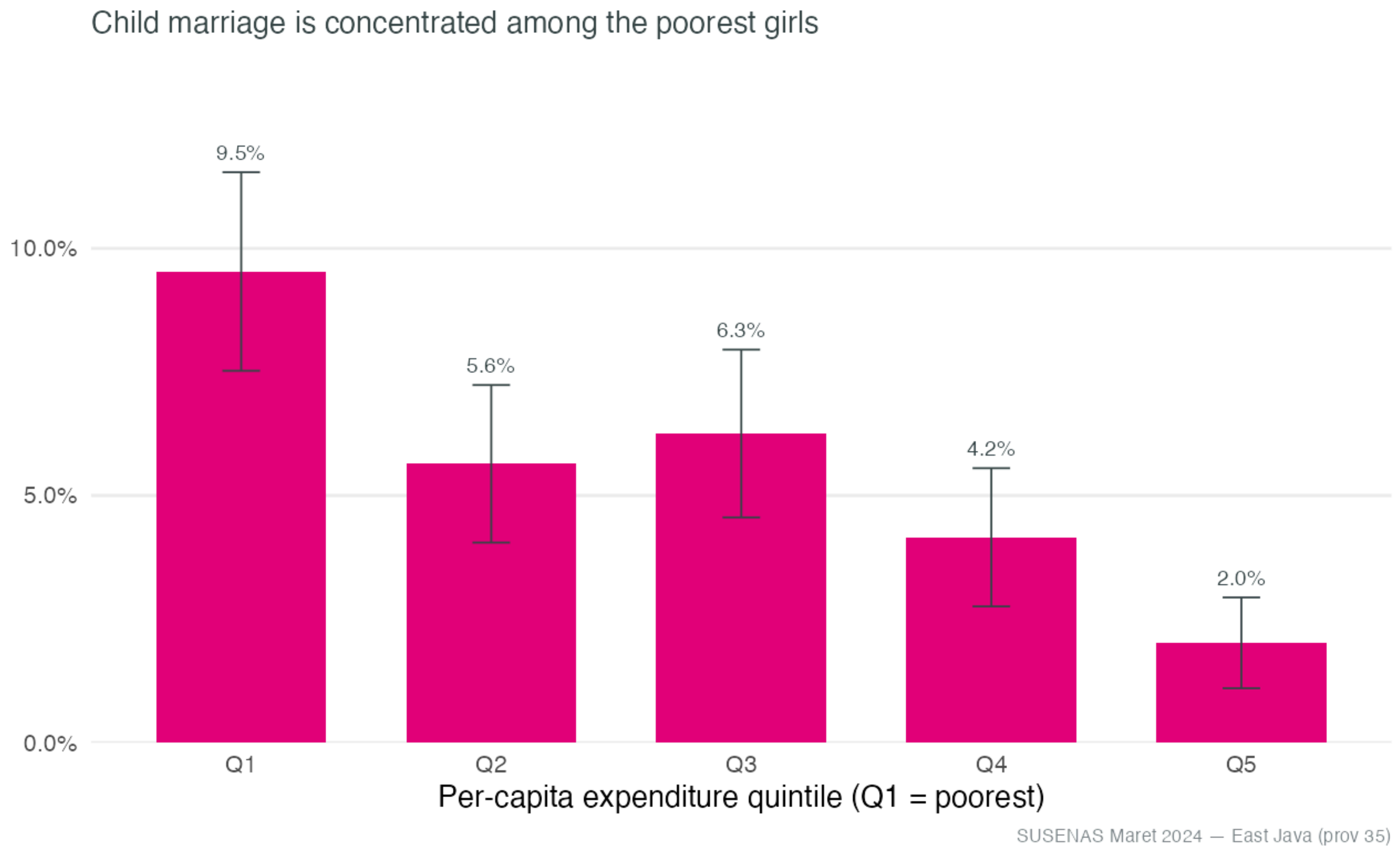
Girls 15–24, SUSENAS Maret 2024, survey-weighted (95% CIs). Girls married before 18 are far more likely to be NEET and out of school. **Association, not causal** — direction is not identified (marriage and leaving school are mutually reinforcing). 5.4% of East-Java girls 15–24 married before 18 (vs 0.4% of boys), from the household roster's age-at-first-marriage question.

Early pregnancy — the twin pathway



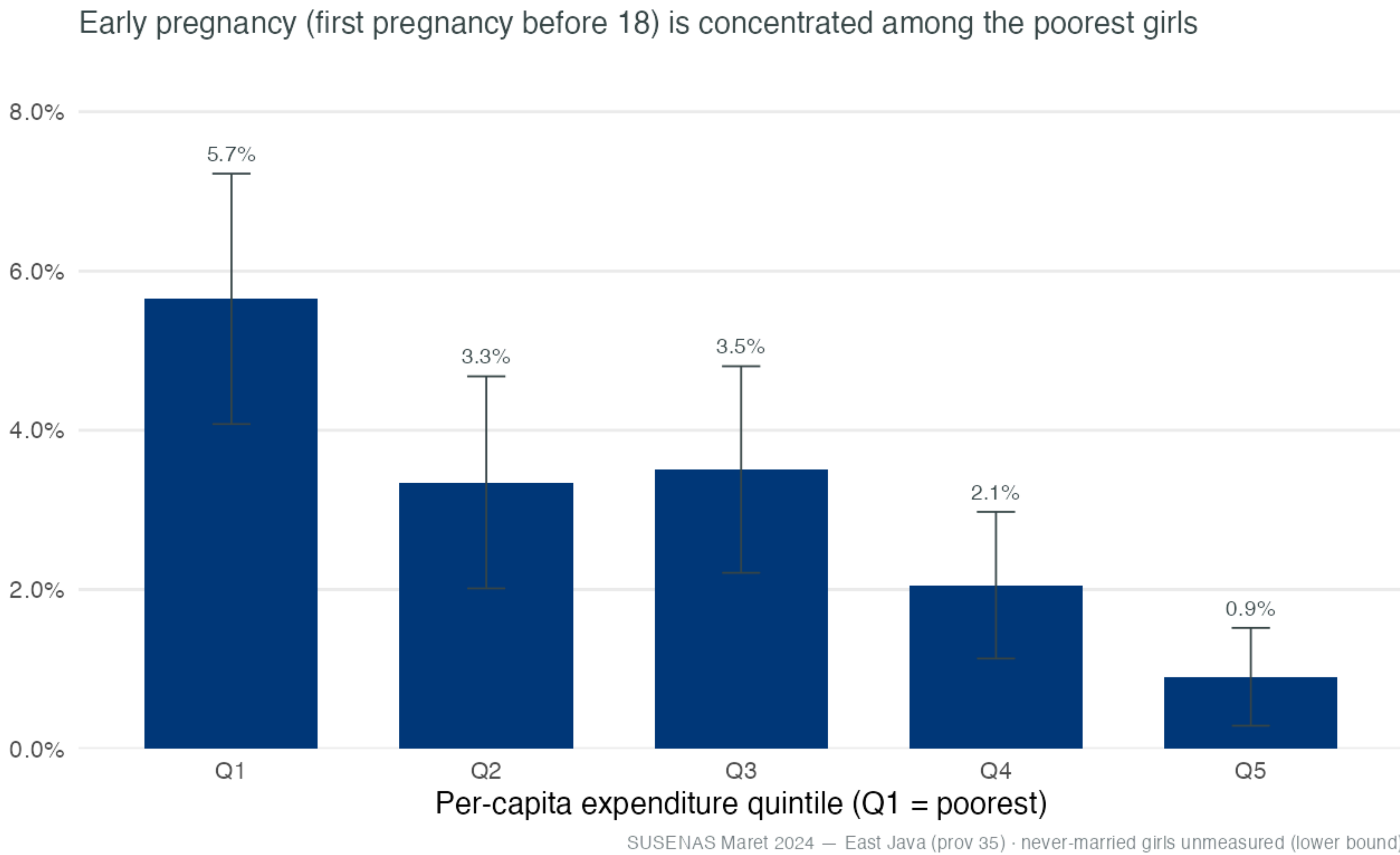
First pregnancy before 18 (SUSENAS Blok XV). Early pregnancy and child marriage are tightly sequenced — pregnancy often precipitates marriage under social pressure. SUSENAS records pregnancies only for ever-married women, so these are **lower-bound** estimates.

Poverty drives child marriage



Child marriage falls steeply from the poorest to the richest expenditure quintile — the mechanism a cash-transfer (Cash+) component is designed to interrupt. Indonesia's own PKH cut girls' child marriage by ~3.5pp (Priebe & Sumarto, 2025).

Poverty also drives early pregnancy



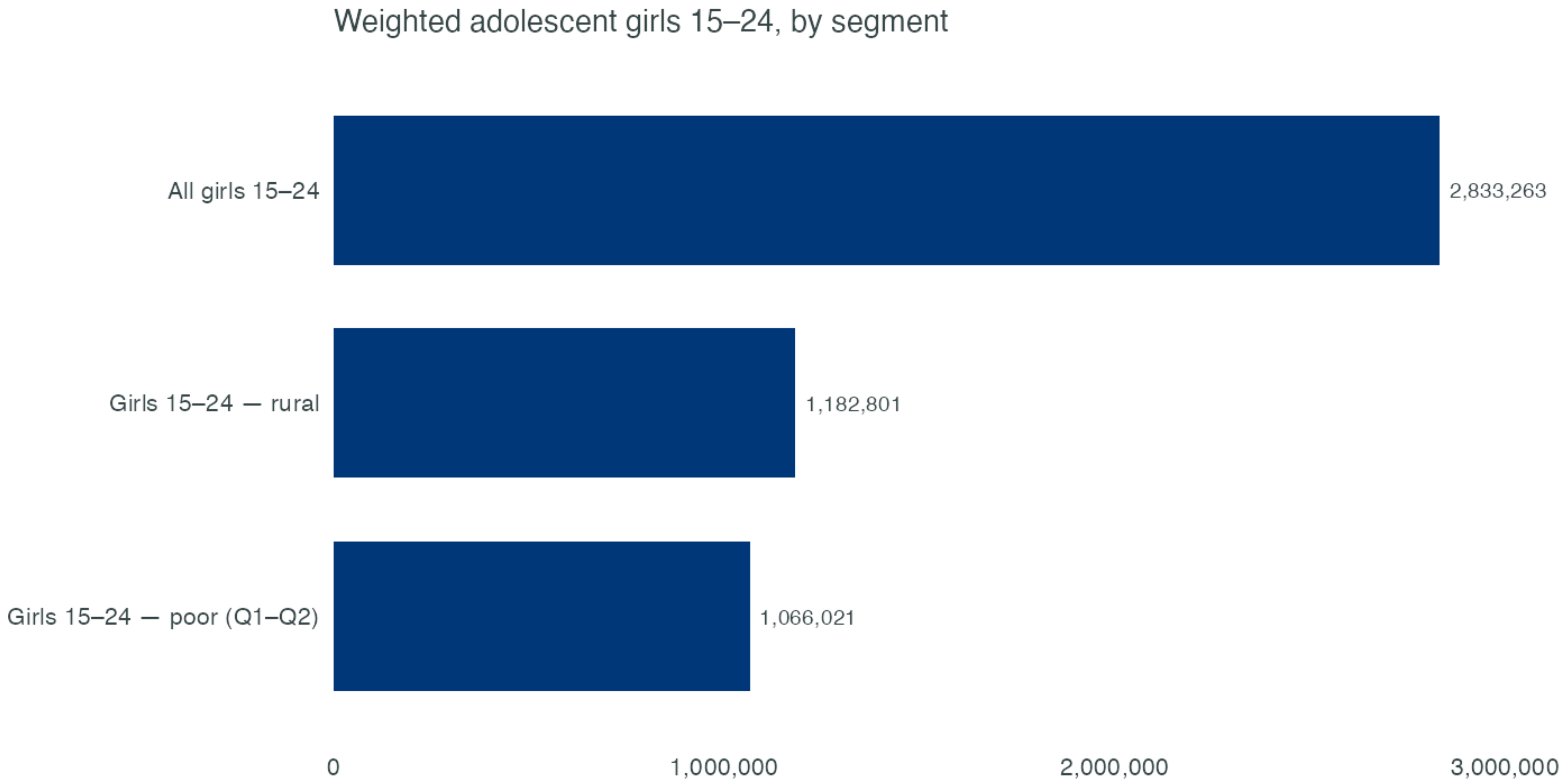
The same poverty gradient as child marriage — one mechanism, two manifestations. This is the entry point for a **Cash+ component**: PKH reduced girls' child marriage by ~3.5pp (Priebe & Sumarto, 2025).

Formal tests — marriage, pregnancy, poverty & exclusion

Hypothesis	Effect (pp)	SE	p
Poverty (Q1–Q2) → child marriage	2.6	0.8	0.0011
Child marriage → NEET	36.5	3.2	0.0000
Child marriage → in school	-29.8	2.4	0.0000
Child marriage → employment	-12.0	3.1	0.0001
Poverty × child marriage (NEET amplification)	-8.7	6.2	0.1580
Poverty (Q1–Q2) → early pregnancy	2.1	0.6	0.0009
Early pregnancy → NEET	36.7	4.2	0.0000
Early pregnancy → in school	-29.3	3.0	0.0000
Early pregnancy → employment	-12.6	3.6	0.0006
Early pregnancy ↔ child marriage (association)	95.9	0.5	0.0000
Poverty × early pregnancy (NEET amplification)	-3.7	8.3	0.6510

Linear-probability estimates, girls 15–24, survey-weighted, PSU-clustered, N ≈ 6,100–6,200 per model (SUSENAS Maret 2024). Effects in percentage points. **Controls**: models of marriage/pregnancy → outcomes include age, urban/rural, and per-capita-expenditure poverty (Q1–Q2); models of poverty → marriage/pregnancy, and the two interaction terms, include age and urban/rural. Robust to adding **district fixed effects**. Poverty raises the risk of both marriage and pregnancy, and marriage/pregnancy independently predict much higher NEET — but the **poverty × marriage/pregnancy interaction is not statistically significant** (p=0.16 and p=0.65), so the data do not support an amplified penalty specifically for poor girls; the case for pairing skills programming with cash transfers rests on the poverty→marriage/pregnancy pathway itself, reinforced by PKH's own child-marriage evidence. Association, not causal identification.

Target population — adolescent girls 15–24



Source: SUSENAS Maret 2024 — East Java (prov 35). Weighted, adolescents 15–24.

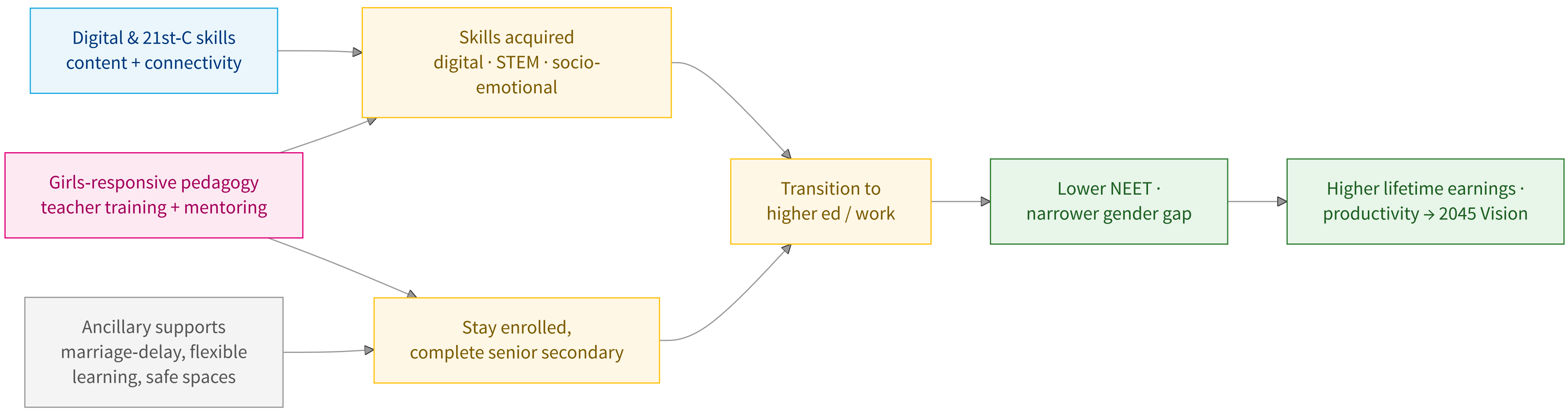
2 · Evidence → Theory of Change

What works — best-matched evidence (raw)

Programme	Design	Girls / age	Focus	Key result	Fit for East Java
Adolescent girls' wellbeing — Bergström & Özler 2021	Review (104 studies)	Adolescent girls	Multi-sector interventions	Integrated packages beat single-sector ones	Synthesis backbone for the ToC
NEET in Indonesia — Faizah, Isbah & Azca 2025	Qualitative	Youth (gendered)	Drivers of girls' NEET	Non-education factors (norms, care, marriage) drive girls' NEET	Indonesia-specific NEET context
ELA girls' clubs — Bandiera 2018 (Sierra Leone)	RCT	Girls 12–25	Vocational + life skills	Enrolment +8.5pp; more income-generating activity	Closest Skills4Girls-in-Double-Track model
Female digital-employment bootcamp — Alzúa 2020 (Argentina/Colombia)	RCT	Women 18–50	Coding for employment	Large coding-skill gains	Core digital-skills → jobs arm
Irumi — Näslund-Hadley 2025 (Paraguay)	RCT	Pupils	Computational thinking	+0.09 SD CT	Structured CT pedagogy
Robotics & Coding — Gülen 2025 (Turkey)	RCT	Girls 14–16	Robotics / coding	Higher STEM-career orientation	Girls' STEM, secondary age
Samata — Prakash 2019 (India)	RCT	Poor girls 13–16	Secondary-transition support	More girls transition; norm shift	South Asia; senior-secondary transition
Room to Read GEP — Edmonds 2021 (India)	RCT	Girls	Life skills + mentoring	Dropout −3.5 to −4.1pp	Mentoring/pedagogy layer cuts dropout
Negotiating a Better Future — Ashraf 2025 (Zambia)	RCT	Girls ~14	Negotiation skills	+0.26 years schooling (10-yr)	Durable, cheap soft-skills layer
OLPC / Plan Ceibal — de Melo 2014 (Uruguay)	Quasi / review	Pupils	Laptops (hardware-led)	Learning ≈ null (−0.05 SD)	Caution: hardware alone fails → fund pedagogy

Raw effects, **as reported in each source** — not yet anchored. Source: project evidence database, 51 studies (Grueso et al., 2026, paper in development). No Indonesia RCT exists, so effects are ϕ-transported to East Java (method in Section 5 · Methods & Data).

Theory of change — skills for girls → lower NEET



Embedded in Double Track / **Tambahan Keterampilan**. Evidence: ELA → O2/M1; Alzúa/Irumi/Gülen → O1/M1; Room to Read/Negotiating → O2 (pedagogy layer); OLPC → caution on hardware-led A1. Measured against the ToR: cost per NEET averted, transition to HE/work, BCR/NPV.

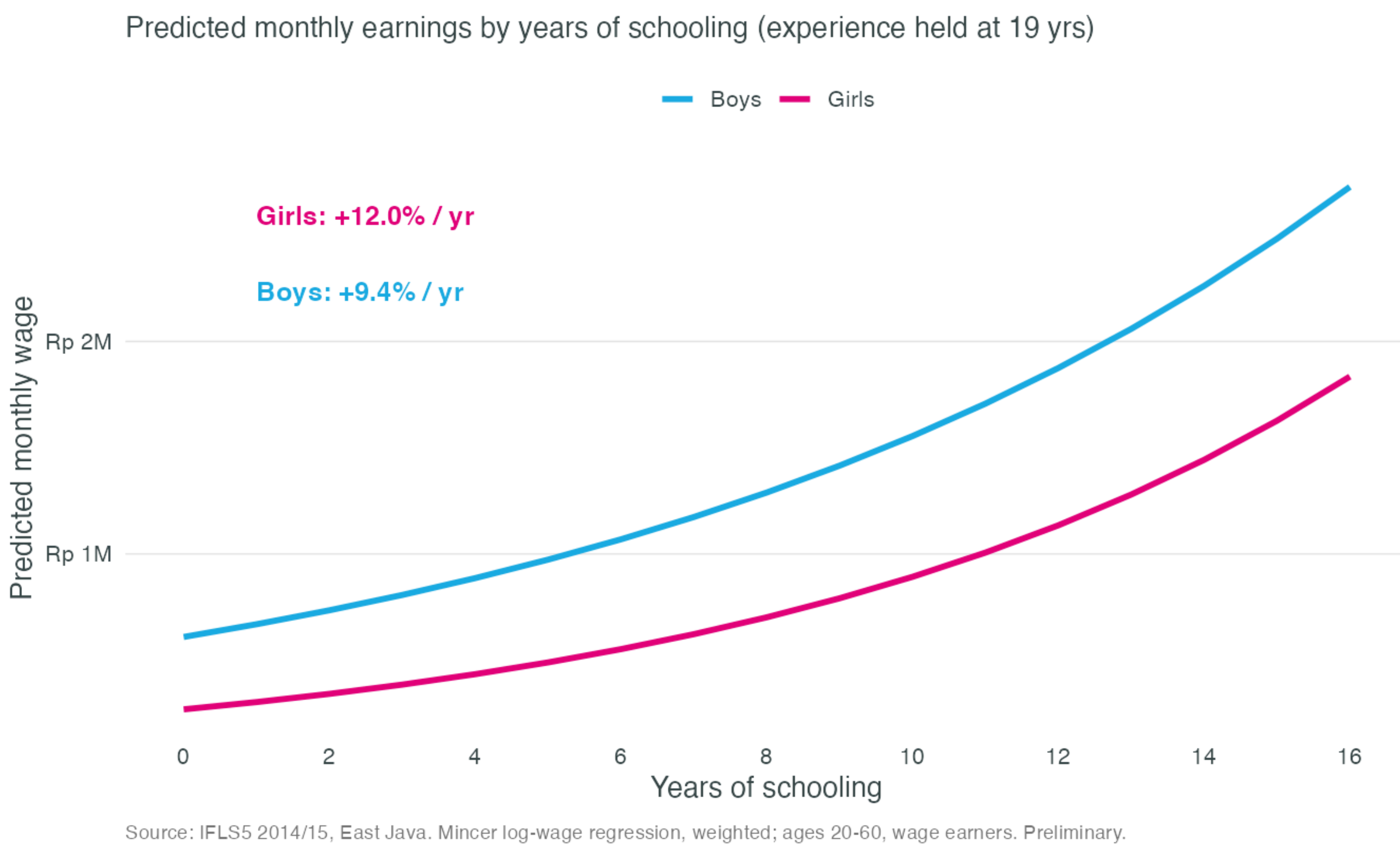
Raw effect → anchored effect

Programme	Raw effect (95% CI)	ϕpop	ϕgeo	ϕecon	ϕdeliv	Adjusted (95% CI)	Sig
ELA girls' clubs (Bandiera 2018) — SLE	+0.085 [-0.011, +0.181] pp	0.95	0.65	0.80	0.70	+0.029 [-0.004, +0.063]	—
Female coding bootcamp (Alzúa 2020) — Arg./Col.	+1.080 [+0.719, +1.441] score-SD	0.70	0.50	0.85	0.70	+0.225 [+0.150, +0.300]	✓
Negotiation skills (Ashraf 2025) — ZMB	+0.264 [+0.041, +0.487] yrs	0.90	0.65	0.80	0.70	+0.086 [+0.013, +0.160]	✓
Room to Read GEP (Edmonds 2021) — IND	-0.035 [-0.070, +0.000] pp dropout	0.95	0.85	0.90	0.70	-0.018 [-0.036, +0.000]	—
Irumi comp. thinking (Näslund-Hadley 2025) — PRY	+0.089 [-0.013, +0.191] SD	0.80	0.50	0.85	0.70	+0.021 [-0.003, +0.045]	—

5a2 **Illustrative** ϕ **(judgment bands), not yet calibrated** — to be set against East-Java baselines from SUSENAS Maret 2024. Each row keeps its own outcome unit. Sig. ✓ = adjusted CI excludes zero. Raw effects/SEs from the evidence database (Grueso et al., 2026).

3 · Projections — East Java

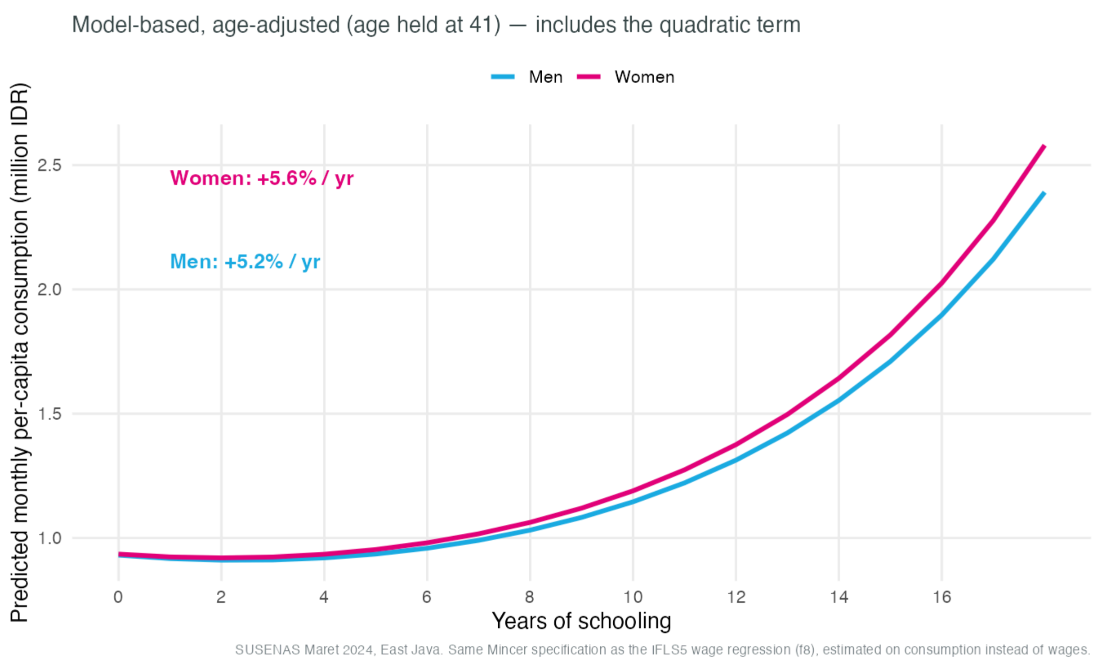
Returns to schooling — girls vs boys (wage-based)



Earnings are **not collected in SUSENAS KOR**, so the Mincer return is estimated on **IFLS5 (2014/15)** wage earners 20–60 and applied to East Java. Raw return; girls +12.0%/yr, boys +9.4%/yr — entered into the cost-benefit at a selection-bounded **7–12%** range. Refine with Sakernas East-Java wages.

Predicted consumption by schooling — girls vs boys

Since SUSENAS collects no wages, we estimate the **welfare return to schooling** using household **per-capita consumption** (`KAPITA`) — adults 25–54, survey-weighted, PSU-clustered, same quadratic specification as the Mincer equation below, so the curve is **model-based and age-adjusted**.



Fitted values from the by-sex regression, age held at the sample mean — girls' return is again slightly above boys', consistent with the wage-based estimate.

Returns to education — consumption-based (SUSENAS 2024), the estimates

Model	Term	Estimate	SE	p
Overall (linear)	years_school	0.054	0.002	0.000
Overall (quadratic)	years_school	-0.018	0.004	0.000
Overall (quadratic)	years_school^2	0.004	0.000	0.000
Women	years_school	0.056	0.002	0.000
Men	years_school	0.052	0.002	0.000
Women (quadratic)	years_school	-0.016	0.005	0.000
Women (quadratic)	years_school^2	0.004	0.000	0.000
Men (quadratic)	years_school	-0.019	0.005	0.000
Men (quadratic)	years_school^2	0.004	0.000	0.000

Coefficient on years of schooling ≈ proportional gain in per-capita consumption per year. Household-level consumption with individual education (a welfare, not wage, return). A **positive** `years_school²` term indicates **convex** (increasing) returns, shown separately by sex above.

4 · Costing & ROI

Component costs — pedagogy-first, sourced & ranged

All figures are **USD per girl, per year of programme exposure**.

USD per girl per year — Low / Central / High

Component	Low	Central	High
Digital content & platform	5	10	15
Teacher training (annualised)	8	15	25
Mentoring & socio-emotional	5	10	15
School connectivity/devices (shared)	8	15	25
Ancillary: marriage-delay + flexible learning	4	8	14
TOTAL per girl/year (USD)	30	58	94

Source/method: Programme- and literature-benchmarked unit costs, using Indonesia-specific sources where available. Training and mentoring costs are annualised from comparable programmes; connectivity assumes shared labs rather than 1:1 devices. Ancillary activities reflect ToR guidance, and overheads follow UNICEF's 10–20% range. Low/central/high estimates capture sourcing uncertainty. Replace with Double Track actuals once provincial financing data are available.

Benefit value chain

ϕ-adjusted NEET reduction (pp) → Extra years of senior-secondary schooling → **Female Mincer return** IFLS5 wage / SUSENAS consumption → × wage × LFP × working life

Benefit, PV per girl ÷ Cost = Σ components × exposure × (1+overhead) = BCR

The effect chain (top row) runs left to right into the benefit; cost is computed independently, and the two combine into the BCR.

BCR & cost per NEET averted — a range, not a point

Scenario	Years edu gained	Benefit PV (USD/girl)	Cost (USD/girl)	BCR	Cost per NEET averted (USD)
Low	0.04	13	451	0.03	22,560
Central	0.15	233	200	1.17	4,002
High	0.32	1197	66	18.13	825

Honest bounding: the **Low** scenario carries the ϕ-adjusted effect's null (CI includes zero), so its BCR/cost-per-NEET is undefined — the programme's return is conditional on a real effect, to be confirmed at pilot. Every input is a sourced, editable assumption (see Methods & Data).

What the BCR does and does not count

✓ Counted

Higher schooling → lifetime earnings premium for girls moved out of NEET, via the female Mincer return (wage or consumption based) applied over the discounted working life.

— **Not counted** (so the BCR is conservative)

Intergenerational gains (Akresh et al. 2022) · delayed child marriage & reduced IPV · health & productivity spillovers · fiscal returns · broader empowerment

Poverty enters as **long-term productivity** / **lifetime earnings**, not immediate income transfers (MoF-defensible framing, per ToR).

5 · Key Findings

Key findings

1. The gap is in the transition, not the classroom. Girls match or lead boys in schooling, but carry ~2× the NEET rate (28% vs 13%) and far lower employment.

2. Child marriage and early pregnancy are major, measurable, and tightly linked exclusion pathways. 5.4% of girls 15–24 married before 18 (vs 0.4% of boys); those girls are far more likely to be NEET (75% vs 25%) and out of school. Early pregnancy shows the identical pattern (76% NEET vs 26%; lower-bound, ever-married women only) and is itself strongly associated with child marriage.

3. Both pathways are poverty-graded. Child marriage and early pregnancy both fall steeply from the poorest to the richest expenditure quintile, and poverty significantly raises the risk of each (formal tests, $p<0.01$) — the empirical basis for pairing skills programming with a **Cash+ (cash-transfer) component**, reinforced by Indonesia's own PKH evidence (−3.5pp child marriage).

4. The data do not show a significantly larger penalty for poor girls specifically — the poverty × marriage/pregnancy interaction is not statistically significant. The cash-transfer rationale rests on poverty driving marriage/pregnancy risk itself, not on an amplified exclusion penalty.

5. Returns to schooling are real, convex, and slightly higher for girls across two independent measures — wages (IFLS5, +12.0%/yr) and household consumption (SUSENAS 2024, +5.6%/yr) — with the pay-off steepening at senior-secondary and beyond.

6. Disability cannot currently be assessed. The Washington Group items are absent from the SUSENAS extract in hand; this is flagged as a data gap, not filled with a guess.

6 · Methods & Data

Data & sample — current source

Source. SUSENAS Maret 2024, East Java (province 35) — KOR individual file (`ssn202403_kor_ind1`, 104,599 persons), with per-capita monthly expenditure (`KAPITA`, KP block 43) merged by household. Adolescents 15–24: **13,340** (6,221 girls). **Weighted, PSU/strata-clustered** (`FWT`, `PSU`, `STRATA`) → exact standard errors.

Key constructions:

- **NEET (15–24)** = main activity (`R704`) is neither **working** nor **schooling**, and not currently attending (`R610`). Activity-based measure; aligns with BPS practice. (`R701`/`R702` were found to invert the gender pattern and are **not** used.)
- **Sex** `R405` (1 = male, 2 = female) · **age** `R407` · **married before 18** from age at first marriage (`R409`) · **early pregnancy** from age at first pregnancy (Blok XV, ever-married women only).
- **Wealth quintiles** from per-capita expenditure (`KAPITA`).

Validation. Weighted girls 15–24 = **2.83M** (ToR ≈ 2.77M); girls NEET ≈ **30%** (matches ToR 30.5%) and **higher than boys' ≈ 16%** — the expected gender pattern.

Still to pin (from BPS codebook): internet/digital use (confirmed usable); disability (Washington Group) — see contingency plan.

Data availability — contingency plan

Indicator	Preferred source	Status	Fallback
NEET, schooling, employment	SUSENAS 2024 (R610/R704)	**Available**	—
Child marriage (<18)	SUSENAS R409	**Available**	—
Early pregnancy/birth (<18)	SUSENAS Blok XV (ind2)	**Available** — ever-married only (lower bound)	IDHS 2022 for all-women rates
Consumption / wealth quintiles	SUSENAS KAPITA	**Available**	—
Wages	Not collected in SUSENAS	Consumption Mincer used (5.4–5.6%/yr, convex)	Sakernas wage Mincer; IFLS5 (12%)
Disability (Washington Group)	SUSENAS Blok X (R1002–R1009)	**Not in current extract** — confirmed absent after checking `ind1`, `ind2`, and the merged household file	Re-export KOR keeping Blok X; else BPS published tables
Programme costs	Double Track budget (Prov. Govt)	Pending from client	Literature/EU-modelling unit costs (see Costing & ROI)

Per ToR: definitions of NEET/employment to be validated with BPS.

Transporting evidence to East Java — the ϕ method

No Indonesia RCT exists, so each raw effect is **attenuated** to East Java by a transparent, multiplicative factor (Grueso et al., 2026):

$$\text{effect}_{\text{EJ}} = \text{effect}_{\text{raw}} \times \phi_{\text{pop}} \times \phi_{\text{geo}} \times \phi_{\text{econ}} \times \phi_{\text{deliv}}$$

- ϕ_{pop} — match on sex/age/rural (girls-adolescent anchors score high)
- ϕ_{geo} — distance band (South/SE Asia > SSA / LAC > Europe)
- ϕ_{econ} — economic similarity, capped [0.30, 1.00]
- ϕ_{deliv} — government roll-out vs NGO/research trial

Each $\phi \leq 1$ → **only attenuates, never amplifies**; 95% CIs propagated; the null stays in the low scenario.

The Mincer specification, formally

Both the wage-based (IFLS5) and consumption-based (SUSENAS 2024) returns to schooling are estimated from the same underlying specification, run separately by sex:

$$\ln(Y_i) = \beta_0 + \beta_1 S_i + \beta_2 S_i^2 + \beta_3 \mathrm{Age}_i + \beta_4 \mathrm{Age}_i^2 + \beta_5 \mathrm{Urban}_i + \varepsilon_i$$

- Y_i — monthly wage (IFLS5 earners 20–60) **or** household per-capita expenditure (SUSENAS, adults 25–54)
- S_i — years of schooling (highest certificate attained)
- β_1 — the linear return per additional year of schooling (the headline %/yr figure)
- β_2 — the quadratic term; $\beta_2 > 0$ **and significant ⇒ convex returns** (pay-off accelerates at higher schooling levels)
- Estimated **weighted**, with **PSU-clustered** standard errors (SUSENAS) or robust SEs (IFLS5)